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An Experiment Sequence Involving Potassium Trioxalatoferrate (III) Trihydrate

When choosing experimental sequences for first-year chemistry laboratory, we try to make selections involving a theme of two to four weeks duration reminiscent of research or industrial laboratory work. The synthesis, analysis, and photochemical reactions of potassium trioxalatoferrate (III) trihydrate ($\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ or FeOx_3) is such a sequence. The compound may be synthesized by metathetical reaction,¹ or by hydrogen peroxide oxidation of ferrous oxalate in the presence of excess oxalate.² Facile crystallization to emerald green monoclinic crystals occurs during a one-week period, and many times crystals as large as 0.5 cm³ are isolated. Titration with potassium permanganate can be used to sequentially determine oxalate and then iron with good results.³ Although the trihydrate can be dried over CaCl_2 , quantitative dehydration to the anhydrous compound is possible using P_2O_5 as desiccant. The photochemically active compound is unstable to light and has been used in actinometry.⁴ Aqueous solutions of the compound and $\text{K}_3\text{Fe}(\text{CN})_6$ are useful in making blueprint paper.

Our selections of experiments are also subjected to cost analysis, since we try to spend no more than \$0.20 per student interaction in expendable materials. Conventional permanganate analysis of iron requires reduction to ferrous state by SnCl_2 followed by oxidation of excess SnCl_2 with HgCl_2 . Mercuric chloride is a very expensive chemical. Cost evaluation leads to estimates of \$0.12 to \$0.80 per interaction for the HgCl_2 . Also, we are reluctant to discharge 20 lbs of mercury into our sewage system or to develop a viable mercury recovery procedure. We therefore developed an alternative but less effective procedure that was, nevertheless, consistent with the objectives of the analytical experiment.

Students freshly prepare a 1:10 dilution of KMnO_4 from a 0.1 M stock solution. This solution requires standardization, and must be used quickly. A 0.15 g sample of $\text{Na}_2\text{C}_2\text{O}_4$ is placed in a wide mouth 250-ml titration flask. About 25 ml of water and 10 ml of 3 M H_2SO_4 are added. About 10 ml of 0.01 M KMnO_4 are added from a buret, the flask is heated until the purple color is discharged, and then the warm solution is titrated with the KMnO_4 to a 30-sec faint pink endpoint. Student prepared samples of FeOx_3 are titrated using an identical procedure with 0.20-g samples. At this point the reduction of ferric to ferrous must be accomplished. The titration flask in which oxalate oxidation has taken place is heated to boiling. Additions of small amounts of reagent grade zinc dust are made until the yellow color is discharged. Heating is continued for two minutes beyond this point. The hot mixture is then filtered through Whatman #1 paper into a fresh titration flask. After rinsing to achieve quantitative transfer, a second permanganate titration is performed to a faint pink endpoint. Typical results are shown in the table.

As indicated in the table, the modified procedure gives good agreement between theory and practice. The iron assay is necessarily of high error because of the design of

Permanganate Analysis of Student Prepared $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ Using Zinc Dust Reduction

I. Standardization

Trial	1	2	3
Wt. $\text{Na}_2\text{C}_2\text{O}_4$ (g)	0.1473	0.1412	0.1724
Volume KMnO_4	32.22	30.92	37.51
Calculated Molarity	0.01365	0.01363	0.01372
Average Molarity = $0.01367 \pm 0.00004 M$ (0.3% relative precision)			

II. Analysis

(Estimation of formula weight assuming: (A), Three oxalate moieties; and (B), one iron atom per unit.)

Trial	1	2	3
Wt. sample (g)	0.1778	0.2369	0.1961
(A) Three oxalates/unit			
Volume KMnO_4 (ml)	31.96	42.32	35.17
Calculated Formula			
Weight (g/mole)	488.4	491.4	489.5
Average Formula Weight = $489.8 \pm 1.1 \text{ g/mole}$ (0.2% relative precision) (0.3% absolute error)			
(B) One iron/unit			
Volume KMnO_4 (ml)	5.34	6.99	5.64
Calculated Formula			
Weight (g/mole)	487.1	495.8	508.7
Average Formula Weight = $497 \pm 8 \text{ g/mole}$ (1.5% relative precision) (1.2% absolute error)			

Accepted Formula Weight $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O} = 491.25 \text{ g/mole}$

the experiment. In this case, the error in buret reading is about 0.5% of the measured volume. A more efficient procedure would be to prepare a reduction column using a zinc amalgam prepared with 20–40 mesh zinc metal and a small amount of mercuric chloride. It did not seem to be worth the extra effort to do this because of the large built-in error with the experiment's design.

We use the H_2O_2 synthetic route during the first experimental session. Crystal isolation and blueprinting are accomplished during the second session. Ground samples of compound are stored over CaCl_2 between the second and third sessions, and permanganate titration analysis is accomplished during the third session. It is a good idea to have an extra supply of FeOx_3 on hand during the third session for those students who, for one reason or another, do not have satisfactory analytical samples.

We would like to acknowledge helpful discussions with Drs. Helen Brooks, Robert Levenson, Fred Sicilio, and Roger Whealy.

¹ Bailar, Jr. J. C., and Jones, E. M., "Inorganic Syntheses," McGraw-Hill, Inc., New York, 1939, Vol. I, p. 36.

² Sienko, M. J., and Plane, R. A., "Experimental Chemistry," (4th Ed.), McGraw-Hill, Inc., New York, 1972, p. 165.

³ Trefonas, L. M., and Tucker, Jr., W. C., "Analysis and Equilibrium. Volume 2. Integrated Laboratory Sequence." (Prelim. Ed.), Barnes and Noble, Inc., New York, 1967, p. 122.

⁴ Hatchard, C. G., and Parker, C. A., *Proc. Roy. Soc. (London)*, 235, 518 (1956).